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Trigonometry and coordinates

Level 3 Senior · Geometry

Use the sine and cosine rules, the area formula, coordinates and the shoelace formula to find lengths, angles and areas quickly.

By the end you can: choose between the sine rule, the cosine rule and the area formula to find lengths, angles and areas; set up coordinates sensibly and find areas with the shoelace formula; decide quickly whether a problem is best attacked synthetically, with trigonometry or with coordinates.

Lesson

A triangle has sides 5 and 8 with an angle of 60° between them. To find the third side, the area and the radius of the circumcircle, use the cosine rule for the third side, $\sqrt{25+64-40}=7$, the area formula for $\frac{1}{2}\cdot 5\cdot 8\sin 60^\circ = 10\sqrt{3}$, and the extended sine rule for $R=\dfrac{7}{2\sin 60^\circ}=\dfrac{7}{\sqrt{3}}$. Doing the same with altitudes takes much longer. This unit covers these formulas, coordinates and the shoelace formula, with the aim that you can choose between them quickly and recognise when a synthetic argument is shorter.

Area: one half $ab\sin C$

Area formula. In any triangle ABC, writing $a=BC$, $b=CA$ and $[ABC]$ for the area,

$$[ABC]=\tfrac{1}{2}ab\sin C.$$

Proof: the height from A to the line BC is $b\sin C$. If C is obtuse, the height is $b\sin(180^\circ-C)$, which is the same number. Now use half base times height.

In short-answer contests the most useful consequence involves no trigonometry. If X lies on AB and Y on AC, then triangles AXY and ABC share the angle A, so

$$\frac{[AXY]}{[ABC]}=\frac{\frac{1}{2}\,AX\cdot AY\sin A}{\frac{1}{2}\,AB\cdot AC\sin A}=\frac{AX\cdot AY}{AB\cdot AC}.$$

Two triangles with a common angle, or with supplementary angles since $\sin(180^\circ-\theta)=\sin\theta$, have areas in the ratio of the products of the sides around that angle.

Example 1. Triangle ABC has area 60, $AB=10$ and $AC=15$. Points X on AB and Y on AC satisfy $AX=5$ and $AY=9$. Find the area of the quadrilateral $XBCY$.

Solution.
One could find the angle first: $60=\frac{1}{2}\cdot 10\cdot 15\sin A$ gives $\sin A=\frac{4}{5}$, and then compute heights and subtract. The ratio formula is shorter:

$$[AXY]=60\cdot\frac{5\cdot 9}{10\cdot 15}=18,$$

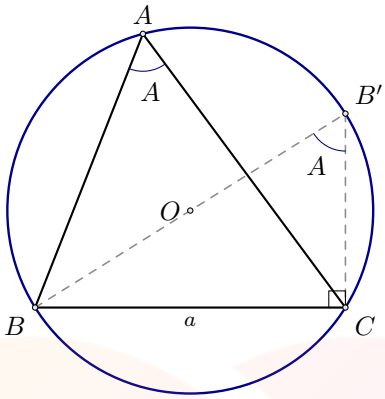
so $[XBCY]=60-18=42$. Only the fact that the angle is shared was used, not its size.

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The sine rule and the circumradius

Extended sine rule. In triangle ABC with circumradius R,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R.$$



Proof: let O be the circumcentre and draw the diameter BB'. If A is acute, then A and B' lie on the same arc cut off by BC, so $\angle BB'C = \angle BAC$ (angles in the same segment), while $\angle BCB' = 90°$ (angle in a semicircle). In the right-angled triangle BCB' we get $a = BC = BB' \sin A = 2R \sin A$. If A is obtuse, B' lies on the other arc, so $\angle BB'C = 180° - A$, which has the same sine; if $A = 90°$, then a is itself a diameter. The other two ratios follow in the same way.

Combining this with the area formula gives the identity $[ABC] = \frac{1}{2}ab \sin C = \dfrac{abc}{4R}$.

One warning. If you know two sides and an angle that is *not* between them, there may be two different triangles. Since $\sin\theta = \sin(180° - \theta)$, the sine rule cannot tell an acute angle from its obtuse partner, and you have to decide which one (or both) is possible.

The cosine rule

Cosine rule. In any triangle, $c^2 = a^2 + b^2 - 2ab\cos C$.

The following proof uses coordinates, which are treated below. Put C at the origin and CA along the positive x-axis, so that $A = (b, 0)$ and $B = (a\cos C,\ a\sin C)$. Then

$$c^2 = AB^2 = (a\cos C - b)^2 + a^2\sin^2 C = a^2(\cos^2 C + \sin^2 C) - 2ab\cos C + b^2,$$

which is the rule. It works for obtuse C too, where $\cos C < 0$ and the third side is longer than Pythagoras would give. Use it forwards (two sides and the included angle give the third side) or backwards (three sides give any angle). Used backwards on integer sides, it often finds an angle of $60°$ or $120°$.

Example 2. Triangle ABC has $AB = 8$, $BC = 7$ and $CA = 5$. Find its area, its circumradius R, and the distance from the circumcentre O to the side BC.

Solution.
Heron's formula would give the area, but it would not help with the other two parts. Integer sides often come with an angle of $60°$ or $120°$, so we try the cosine rule backwards, starting

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with the angle opposite the middle side:

$$\cos A = \frac{8^2 + 5^2 - 7^2}{2 \cdot 8 \cdot 5} = \frac{40}{80} = \frac{1}{2},$$

so $A = 60°$. Then $[ABC] = \frac{1}{2} \cdot 8 \cdot 5 \sin 60° = 10\sqrt{3}$, and $R = \dfrac{7}{2 \sin 60°} = \dfrac{7}{\sqrt{3}} = \dfrac{7\sqrt{3}}{3}$. For the distance, the angle at the centre is twice the angle at the circumference, so $\angle BOC = 120°$. Triangle BOC is isosceles with $OB = OC = R$, so the perpendicular from O meets BC at its midpoint M and $\angle BOM = 60°$. Hence $OM = R \cos 60° = \dfrac{7\sqrt{3}}{6}$. Check with Pythagoras: $OM^2 + BM^2 = \frac{49}{12} + \frac{49}{4} = \frac{49}{3} = R^2$.

Coordinates

Coordinates replace geometry by algebra, and they are worth using when the algebra will be short. Almost all of the skill is in choosing the axes:

- put the origin at the most important point, often a right angle, a centre, or a vertex with many lines through it;
- put as many lines as possible along the axes, and use symmetry, for example an isosceles triangle with its apex on the y-axis;
- scale freely: if only ratios are asked for, pick side lengths that avoid fractions.

After that you need only a handful of facts: the distance $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$, the midpoint, the gradient $\frac{y_2 - y_1}{x_2 - x_1}$, perpendicular gradients multiplying to -1, and the circle $(x-p)^2 + (y-q)^2 = r^2$.

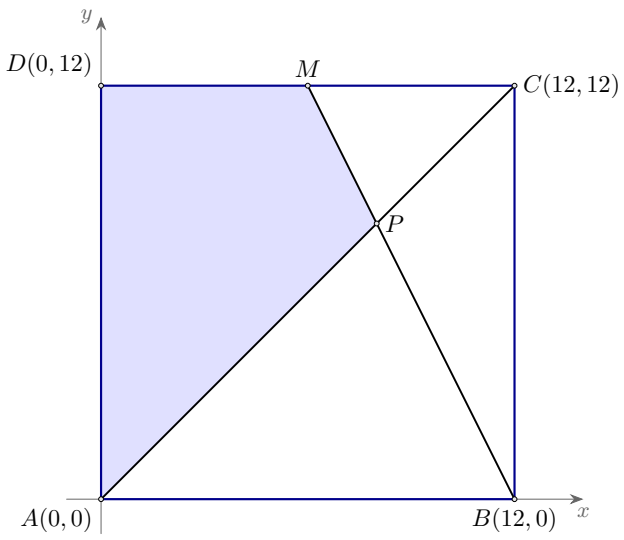
The shoelace formula

Shoelace formula. If a polygon has vertices $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$ listed in order around its boundary, then its area is

$$\tfrac{1}{2} \left| (x_1 y_2 - x_2 y_1) + (x_2 y_3 - x_3 y_2) + \cdots + (x_n y_1 - x_1 y_n) \right|.$$

Proof: for a triangle with vertices $O = (0, 0)$, $P = (x_1, y_1)$ and $Q = (x_2, y_2)$, write $P = (p \cos \alpha, p \sin \alpha)$ and $Q = (q \cos \beta, q \sin \beta)$. The area formula gives $\frac{1}{2} pq \sin(\beta - \alpha)$, and expanding $\sin(\beta - \alpha)$ turns this into $\frac{1}{2}(x_1 y_2 - x_2 y_1)$. So $\frac{1}{2}(x_1 y_2 - x_2 y_1)$ is the area of triangle OPQ, positive when O, P, Q run anticlockwise and negative when clockwise. Add these signed areas over all the edges of the polygon: the parts outside the polygon cancel and you are left with its area, up to sign. In practice, list the vertices in order, repeat the first at the end, and add up the cross products.

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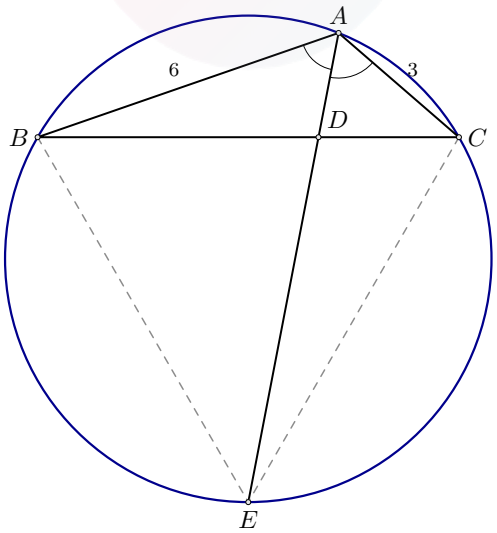


Example 3. Square $ABCD$ has side 12, and M is the midpoint of CD. The segment BM meets the diagonal AC at P. Find the area of the quadrilateral $APMD$.

Solution.
Set $A = (0,0)$, $B = (12,0)$, $C = (12,12)$ and $D = (0,12)$, so $M = (6,12)$. The diagonal is $y = x$. The line BM has gradient $\frac{12-0}{6-12} = -2$, so it is $y = -2x + 24$. They meet where $x = -2x + 24$, at $P = (8,8)$.
Shoelace on $A(0,0)$, $P(8,8)$, $M(6,12)$, $D(0,12)$:

$$\tfrac{1}{2}\,|(0\cdot 8 - 8\cdot 0) + (8\cdot 12 - 6\cdot 8) + (6\cdot 12 - 0\cdot 12) + (0\cdot 0 - 0\cdot 12)| = \tfrac{1}{2}(0 + 48 + 72 + 0) = 60.$$

Compare the synthetic route: $AB \parallel MC$ and $AB = 2\,MC$, so triangles PAB and PCM are similar with ratio 2, and $AP : PC = 2 : 1$. Then $[APMD] = [ACD] - [PCM] = 72 - \frac{1}{3}[ACM] = 72 - \frac{1}{3}\cdot 36 = 60$. Both are quick here. With more lines and intersections, coordinates are usually faster; when the problem reduces to a single ratio, similar triangles are.



Example 4. In triangle ABC, $\angle BAC = 120°$, $AB = 6$ and $AC = 3$. The bisector of angle A meets BC at D and the circumcircle again at E. Find AD and DE.

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Solution.
For AD one could find BC by the cosine rule, split it with the angle bisector theorem, and apply the cosine rule again in triangle ABD, which is three calculations with surds. Areas are shorter (splitting the triangle along the bisector). The bisector splits the 120° angle into two angles of 60°, and $[ABC] = [ABD] + [ADC]$:

$$\tfrac{1}{2} \cdot 6 \cdot 3 \sin 120^\circ = \tfrac{1}{2} \cdot 6 \cdot AD \sin 60^\circ + \tfrac{1}{2} \cdot 3 \cdot AD \sin 60^\circ.$$

All three sines are equal, so $18 = 9\,AD$ and $AD = 2$.
For DE we need the two parts of BC, and then power of a point (from L3-GE1) at D. The cosine rule gives $BC^2 = 36 + 9 - 2 \cdot 6 \cdot 3 \cos 120^\circ = 45 + 18 = 63$, so $BC = 3\sqrt{7}$. The angle bisector theorem gives $BD : DC = AB : AC = 2 : 1$, so $BD = 2\sqrt{7}$ and $DC = \sqrt{7}$. The chords AE and BC meet at D, so

$$AD \cdot DE = BD \cdot DC = 14, \qquad DE = 7.$$

So $AE = 9 = AB + AC$, and this holds whenever $\angle BAC = 120^\circ$. Since AE bisects angle A, the point E is the midpoint of arc BC, so $EB = EC$; and $\angle BEC = 180^\circ - 120^\circ = 60^\circ$ because $ABEC$ is cyclic. So triangle BEC is equilateral, and if you know Ptolemy's theorem it gives $AE \cdot BC = (AB + AC) \cdot BC$ at once. The area argument also generalises: a bisector of length d from an angle A satisfies $\dfrac{2\cos\frac{A}{2}}{d} = \dfrac{1}{b} + \dfrac{1}{c}$, which for $A = 120^\circ$ is simply $\dfrac{1}{d} = \dfrac{1}{b} + \dfrac{1}{c}$.

Common traps.

- The sine rule cannot distinguish θ from $180^\circ - \theta$. With two sides and a non-included angle, check whether one triangle or two exist.
- The cosine of an obtuse angle is negative, so $-2ab\cos C$ becomes a plus: with $C = 120^\circ$, $c^2 = a^2 + b^2 + ab$.
- The SMC is calculator-free. Know the sines and cosines of 30°, 45°, 60° and their supplements cold, and keep answers exact.
- Shoelace needs the vertices in order around the polygon. Listing them in a crossing order gives a wrong number with no warning. Remember the absolute value and the half.
- Bad axes make coordinates painful. If your first line equation already has three fractions in it, stop and choose the origin again.

How to spot this in a contest

The choice of method usually depends on what is given:

1. Two sides and the angle between them: the area formula or the cosine rule.
2. Angles and one side, or any mention of the circumradius: the sine rule.
3. Three sides, when you want an angle or an area: the cosine rule backwards, looking for an angle of 60° or 120°.
4. Squares, rectangles, grids, right angles and constructions of the form "the line through this point meets that line at": coordinates, and shoelace for any polygon whose vertices you can write down.
5. Two triangles sharing an angle: a ratio of areas, with no angle needed.

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Before committing to a long calculation, spend thirty seconds checking whether an angle bisector, a pair of similar triangles or power of a point gives the answer directly.

Practice

Try each problem before turning to the hints and solutions at the back.

1. ⋆ A triangle has two sides of lengths 6 and 10 with an angle of 150° between them. Find its area.

2. ⋆⋆ A triangle has sides of lengths 3, 5 and 7. Find the radius of its circumcircle.

3. ⋆⋆ Rectangle $ABCD$ has $AB = 8$ and $BC = 6$. Point E is the midpoint of BC, and F lies on CD with $DF = 2$. The segments AE and BF meet at P. Find the area of triangle ABP.

4. ⋆⋆⋆ Triangle ABC has area 48. Points D, E, F lie on the sides BC, CA, AB respectively, with $BD : DC = 1 : 3$, $CE : EA = 2 : 1$ and $AF : FB = 1 : 1$. Find the area of triangle DEF.

5. ⋆⋆⋆ In triangle ABC, $AB = 8$, $AC = 7$ and $\angle ABC = 60^\circ$. Find the sum of all possible values of the area of triangle ABC.

6. ⋆⋆⋆⋆ A cyclic quadrilateral $ABCD$ has $AB = 2$, $BC = 3$, $CD = 5$ and $DA = 6$. Find its area.

Contest ladder

Real past problems that use this idea, easiest first. Open them on the official paper or the AoPS wiki, or online at kevinvariant08.github.io.

1. UKMT SMC 2026 Question 3 ⋆
2. UKMT SMC 2023 Question 14 ⋆⋆
3. UKMT SMC 2015 Question 14 ⋆⋆
4. AMC 12A 2020 Problem 12 ⋆⋆
5. UKMT SMC 2021 Question 22 ⋆⋆⋆
6. AIME I 2019 Problem 3 ⋆⋆⋆
7. UKMT SMC 2013 Question 24 ⋆⋆⋆
8. UKMT SMC 2020 Question 24 ⋆⋆⋆
9. AMC 10A 2018 Problem 24 ⋆⋆⋆
10. UKMT SMC 2026 Question 24 ⋆⋆⋆
11. AIME 1990 Problem 7 ⋆⋆⋆
12. AIME I 2001 Problem 4 ⋆⋆⋆⋆

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Hints

1. Use $\frac{1}{2}ab\sin C$, and remember that $\sin 150^\circ = \sin 30^\circ$.
2. Find the largest angle with the cosine rule first. It is a standard angle.
3. Put A at the origin. The area of ABP only needs the height of P above AB.
4. Remove the three corner triangles. Each of them shares an angle with ABC.
5. Call $BC = x$ and use the cosine rule at B. Expect a quadratic, and think about whether both roots count.
6. Write the diagonal AC in two ways with the cosine rule, using the fact that opposite angles are supplementary.

Solutions

Problem 1

Answer: 15

The area is $\frac{1}{2} \cdot 6 \cdot 10 \cdot \sin 150^\circ = 30 \cdot \frac{1}{2} = 15$.

Problem 2

Answer: $\frac{7\sqrt{3}}{3}$

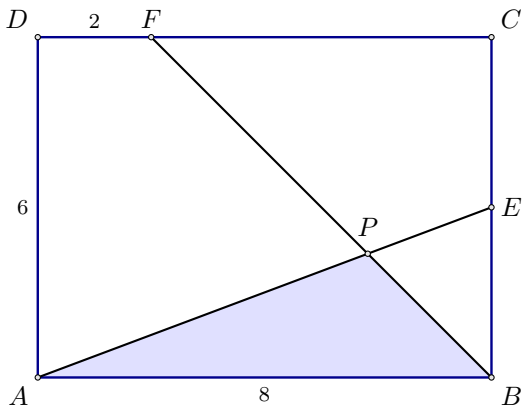
The largest angle θ is opposite the side 7, and the cosine rule gives

$$\cos\theta = \frac{3^2 + 5^2 - 7^2}{2 \cdot 3 \cdot 5} = \frac{-15}{30} = -\frac{1}{2},$$

so $\theta = 120^\circ$. By the extended sine rule, $2R = \dfrac{7}{\sin 120^\circ} = \dfrac{14}{\sqrt{3}}$, so $R = \dfrac{7}{\sqrt{3}} = \dfrac{7\sqrt{3}}{3}$.

Problem 3

Answer: $\frac{96}{11}$



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Take $A=(0,0)$, $B=(8,0)$, $C=(8,6)$ and $D=(0,6)$. Then $E=(8,3)$ and $F=(2,6)$. The line AE is $y=\frac{3}{8}x$. The line BF has gradient $\frac{6-0}{2-8}=-1$, so it is $y=8-x$. Solving $\frac{3}{8}x=8-x$ gives $x=\frac{64}{11}$, and then $y=\frac{24}{11}$.

Triangle ABP has base $AB=8$ and height $\frac{24}{11}$, so its area is $\frac{1}{2}\cdot 8\cdot\frac{24}{11}=\frac{96}{11}$.

Problem 4

Answer: 10

We cut off the three corner triangles (the shared angle ratio). At A: $AF=\frac{1}{2}AB$ and $AE=\frac{1}{3}AC$, so $[AFE]=\frac{1}{2}\cdot\frac{1}{3}\cdot 48=8$. At B: $BF=\frac{1}{2}BA$ and $BD=\frac{1}{4}BC$, so $[BDF]=\frac{1}{8}\cdot 48=6$. At C: $CD=\frac{3}{4}CB$ and $CE=\frac{2}{3}CA$, so $[CDE]=\frac{1}{2}\cdot 48=24$.

Hence $[DEF]=48-8-6-24=10$. No lengths or angles of ABC were needed. When a problem gives only an area and ratios along the sides, this is usually the intended method.

Problem 5

Answer: $16\sqrt{3}$

Let $BC=x$. The cosine rule at B gives

$$49=64+x^2-2\cdot 8\cdot x\cos 60^\circ=64+x^2-8x,$$

so $x^2-8x+15=0$ and $x=3$ or $x=5$. Both roots are positive and each satisfies the cosine rule with the given angle, so both triangles exist: this is the ambiguous case, because the given angle is not between the two given sides.

The area is $\frac{1}{2}\cdot 8\cdot x\sin 60^\circ=2\sqrt{3}\,x$, which is $6\sqrt{3}$ or $10\sqrt{3}$. The sum of the possible areas is $16\sqrt{3}$.

Problem 6

Answer: $6\sqrt{5}$

Opposite angles of a cyclic quadrilateral add to 180°, so $\cos D=-\cos B$ and $\sin D=\sin B$. Compute AC^2 in two ways with the cosine rule. From triangle ABC, $AC^2=4+9-12\cos B$. From triangle ACD, $AC^2=25+36-60\cos D=61+60\cos B$. Equating,

$$13-12\cos B=61+60\cos B,$$

so $\cos B=-\frac{2}{3}$, and then $\sin B=\frac{\sqrt{5}}{3}$ (positive, as B is an angle of a triangle).

The area is $[ABC]+[ACD]=\frac{1}{2}\cdot 2\cdot 3\sin B+\frac{1}{2}\cdot 5\cdot 6\sin D=18\sin B=6\sqrt{5}$.

This agrees with Brahmagupta's formula: with $s=8$, the area is $\sqrt{(s-2)(s-3)(s-5)(s-6)}=\sqrt{180}=6\sqrt{5}$. The cosine rule argument above, carried out with general sides, is the standard proof of that formula.

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